

- [38] Chris Yuhao Liu, Yaxuan Wang, Jeffrey Flanigan, and Yang Liu. Large language model unlearning via embedding-corrupted prompts. In *The Thirty-eighth Annual Conference on Neural Information Processing Systems*, 2024. URL <https://openreview.net/forum?id=e5icsXBD8Q>.
- [39] Yu Li, Han Jiang, Chuanyang Gong, and Zhihua Wei. Destein: Navigating detoxification of language models via universal steering pairs and head-wise activation fusion. In *First Conference on Language Modeling*, 2024. URL <https://openreview.net/forum?id=jq2kNXigPP>.
- [40] Sheng Liu, Haotian Ye, Lei Xing, and James Y. Zou. In-context vectors: Making in context learning more effective and controllable through latent space steering. In Ruslan Salakhutdinov, Zico Kolter, Katherine Heller, Adrian Weller, Nuria Oliver, Jonathan Scarlett, and Felix Berkenkamp, editors, *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, pages 32287–32307. PMLR, 21–27 Jul 2024. URL <https://proceedings.mlr.press/v235/liu24bx.html>.
- [41] Daniel Beaglehole, Adityanarayanan Radhakrishnan, Enric Boix-Adserà, and Mikhail Belkin. Aggregate and conquer: detecting and steering llm concepts by combining nonlinear predictors over multiple layers, 2025. URL <https://arxiv.org/abs/2502.03708>.
- [42] Joris Postmus and Steven Abreu. Steering large language models using conceptors: Improving addition-based activation engineering. *arXiv preprint arXiv:2410.16314*, 2024.
- [43] Ruixuan Huang. Steering llms’ behavior with concept activation vectors, September 2024. Draft manuscript. Available on LessWrong forum.
- [44] Zhuohan Gu, Jiayi Yao, Kuntai Du, and Junchen Jiang. Llmsteer: Improving long-context llm inference by steering attention on reused contexts. *arXiv preprint arXiv:2411.13009*, 2024.
- [45] Bingqing Song, Boran Han, Shuai Zhang, Hao Wang, Haoyang Fang, Bonan Min, Yuyang Wang, and Mingyi Hong. Effectively steer llm to follow preference via building confident directions. *arXiv preprint arXiv:2503.02989*, 2025.
- [46] Varvara Logacheva, Daryna Dementieva, Sergey Ustyantsev, Daniil Moskovskiy, David Dale, Irina Krotova, Nikita Semenov, and Alexander Panchenko. ParaDetox: Detoxification with parallel data. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 6804–6818, 2022. doi: 10.18653/v1/2022.acl-long.469. URL <https://aclanthology.org/2022.acl-long.469/>.
- [47] A. H. Miller, W. Feng, A. Fisch, J. Lu, D. Batra, A. Bordes, D. Parikh, and J. Weston. Parlai: A dialog research software platform. *arXiv preprint arXiv:1705.06476*, 2017.
- [48] Chin-Yew Lin. ROUGE: A package for automatic evaluation of summaries. In *Text Summarization Branches Out*, pages 74–81, Barcelona, Spain, July 2004. Association for Computational Linguistics. URL <https://aclanthology.org/w04-1013/>.
- [49] Chongzhou Fang, Ning Miao, Shaurya Srivastav, Jialin Liu, Ruoyu Zhang, Ruijie Fang, Asmita, Ryan Tsang, Najmeh Nazari, Han Wang, and Houman Homayoun. Large language models for code analysis: do llms really do their job? In *Proceedings of the 33rd USENIX Conference on Security Symposium, SEC ’24, USA*, 2024. USENIX Association. ISBN 978-1-939133-44-1.
- [50] Paul Denny, David H. Smith, Max Fowler, James Prather, Brett A. Becker, and Juho Leinonen. Explaining code with a purpose: An integrated approach for developing code comprehension and prompting skills. In *Proceedings of the 2024 on Innovation and Technology in Computer Science Education V. 1, ITiCSE 2024*, page 283–289, New York, NY, USA, 2024. Association for Computing Machinery. ISBN 9798400706004. doi: 10.1145/3649217.3653587. URL <https://doi.org/10.1145/3649217.3653587>.
- [51] Nalin Wadhwa, Jui Pradhan, Atharv Sonwane, Surya Prakash Sahu, Nagarajan Natarajan, Aditya Kanade, Suresh Parthasarathy, and Sriram Rajamani. Core: Resolving code quality issues using llms. *Proc. ACM Softw. Eng.*, 1(FSE), July 2024. doi: 10.1145/3643762. URL <https://doi.org/10.1145/3643762>.

- [52] Baptiste Roziere, Jonas Gehring, Fabian Gloeckle, Sten Sootla, Itai Gat, Xiaoqing Ellen Tan, Yossi Adi, Jingyu Liu, Romain Sauvestre, Tal Remez, et al. Code llama: Open foundation models for code. *arXiv preprint arXiv:2308.12950*, 2023.
- [53] Yujia Li, David Choi, Junyoung Chung, Nate Kushman, Julian Schrittwieser, Rémi Leblond, Tom Eccles, James Keeling, Felix Gimeno, Agustin Dal Lago, Thomas Hubert, Peter Choy, Cyprien de Masson d’Autume, Igor Babuschkin, Xinyun Chen, Po-Sen Huang, Johannes Welbl, Sven Gowal, Alexey Cherepanov, James Molloy, Daniel J. Mankowitz, Esme Sutherland Robson, Pushmeet Kohli, Nando de Freitas, Koray Kavukcuoglu, and Oriol Vinyals. Competition-level code generation with alphacode. *Science*, 378(6624):1092–1097, 2022. doi: 10.1126/science.abq1158. URL <https://www.science.org/doi/abs/10.1126/science.abq1158>.
- [54] Sahil Bhatia, Jie Qiu, Niranjan Hasabnis, Sanjit A. Seshia, and Alvin Cheung. Verified code transpilation with LLMs. In *The Thirty-eighth Annual Conference on Neural Information Processing Systems*, 2024. URL <https://openreview.net/forum?id=spwE9sLrfg>.
- [55] Anna Rohrbach, Lisa Anne Hendricks, Kaylee Burns, Trevor Darrell, and Kate Saenko. Object hallucination in image captioning. In *Empirical Methods in Natural Language Processing (EMNLP)*, 2018.
- [56] Yifan Li, Yifan Du, Kun Zhou, Jinpeng Wang, Wayne Xin Zhao, and Ji-Rong Wen. Evaluating object hallucination in large vision-language models. In *The 2023 Conference on Empirical Methods in Natural Language Processing*, 2023. URL <https://openreview.net/forum?id=xozJwOkZXF>.
- [57] Haotian Liu, Chunyuan Li, Yuheng Li, and Yong Jae Lee. Improved baselines with visual instruction tuning. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 26296–26306, 2024.
- [58] Wenliang Dai, Junnan Li, DONGXU LI, Anthony Tiong, Junqi Zhao, Weisheng Wang, Boyang Li, Pascale N Fung, and Steven Hoi. Instructblip: Towards general-purpose vision-language models with instruction tuning. In A. Oh, T. Naumann, A. Globerson, K. Saenko, M. Hardt, and S. Levine, editors, *Advances in Neural Information Processing Systems*, volume 36, pages 49250–49267. Curran Associates, Inc., 2023. URL https://proceedings.neurips.cc/paper_files/paper/2023/file/9a6a435e75419a836fe47ab6793623e6-Paper-Conference.pdf.
- [59] Ariana Martino, Michael Iannelli, and Coleen Truong. Knowledge injection to counter large language model (llm) hallucination. In *European Semantic Web Conference*, pages 182–185. Springer, 2023.
- [60] Robert Friel and Atindriyo Sanyal. Chainpoll: A high efficacy method for llm hallucination detection. *arXiv preprint arXiv:2310.18344*, 2023.
- [61] Lei Huang, Weijiang Yu, Weitao Ma, Weihong Zhong, Zhangyin Feng, Haotian Wang, Qianglong Chen, Weihua Peng, Xiaocheng Feng, Bing Qin, et al. A survey on hallucination in large language models: Principles, taxonomy, challenges, and open questions. *ACM Transactions on Information Systems*, 43(2):1–55, 2025.
- [62] Sotirios Panagiotis Chytas, Miso Choi, Hyunwoo J. Kim, and Vikas Singh. Reco: Reminder composition mitigates hallucinations in vision-language models, 2025. URL <https://arxiv.org/abs/2506.22636>.
- [63] Alessandro Favero, Luca Zancato, Matthew Trager, Siddharth Choudhary, Pramuditha Perera, Alessandro Achille, Ashwin Swaminathan, and Stefano Soatto. Multi-modal hallucination control by visual information grounding. In *2024 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024. doi: 10.1109/CVPR52733.2024.01356.
- [64] Shi Liu, Kecheng Zheng, and Wei Chen. Paying more attention to image: A training-free method for alleviating hallucination in lvlms. In Ales Leonardis, Elisa Ricci, Stefan Roth, Olga Russakovsky, Torsten Sattler, and Gül Varol, editors, *Computer Vision - ECCV 2024 - 18th European Conference, Milan, Italy, September 29-October 4, 2024, Proceedings, Part LXXXIII*, volume 15141 of *Lecture Notes in Computer Science*, pages 125–140. Springer, 2024. doi: 10.1007/978-3-031-73010-8_8. URL https://doi.org/10.1007/978-3-031-73010-8_8.

- [65] Sangmin Woo, Donguk Kim, Jaehyuk Jang, Yubin Choi, and Changick Kim. Don’t miss the forest for the trees: Attentional vision calibration for large vision language models. *arXiv preprint arXiv:2405.17820*, 2024.
- [66] Sicong Leng, Hang Zhang, Guanzheng Chen, Xin Li, Shijian Lu, Chunyan Miao, and Lidong Bing. Mitigating object hallucinations in large vision-language models through visual contrastive decoding. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024.
- [67] Lin Long, Rui Wang, Ruixuan Xiao, Junbo Zhao, Xiao Ding, Gang Chen, and Haobo Wang. On LLMs-driven synthetic data generation, curation, and evaluation: A survey. In Lun-Wei Ku, Andre Martins, and Vivek Srikumar, editors, *Findings of the Association for Computational Linguistics: ACL 2024*, pages 11065–11082, Bangkok, Thailand, August 2024. Association for Computational Linguistics. doi: 10.18653/v1/2024.findings-acl.658. URL <https://aclanthology.org/2024.findings-acl.658/>.
- [68] Giulia DeSalvo, Jean-Francis Kary, Lazaros Karydas, Afshin Rostamizadeh, and Sanjiv Kumar. No more hard prompts: Softsr prompting for synthetic data generation. *arXiv preprint arXiv:2410.16534*, 2024.
- [69] Saumya Gandhi, Ritu Gala, Vijay Viswanathan, Tongshuang Wu, and Graham Neubig. Better synthetic data by retrieving and transforming existing datasets. In Lun-Wei Ku, Andre Martins, and Vivek Srikumar, editors, *Findings of the Association for Computational Linguistics: ACL 2024*, pages 6453–6466, Bangkok, Thailand, August 2024. Association for Computational Linguistics. doi: 10.18653/v1/2024.findings-acl.385. URL <https://aclanthology.org/2024.findings-acl.385/>.
- [70] Ruibo Liu, Jerry Wei, Fangyu Liu, Chenglei Si, Yanzhe Zhang, Jinmeng Rao, Steven Zheng, Daiyi Peng, Diyi Yang, Denny Zhou, and Andrew M. Dai. Best practices and lessons learned on synthetic data. In *First Conference on Language Modeling*, 2024. URL <https://openreview.net/forum?id=0JaWBhh61C>.
- [71] Christopher Clark, Kenton Lee, Ming-Wei Chang, Tom Kwiatkowski, Michael Collins, and Kristina Toutanova. Boolq: Exploring the surprising difficulty of natural yes/no questions. In *NAACL*, 2019.
- [72] Xiang Zhang, Junbo Jake Zhao, and Yann LeCun. Character-level convolutional networks for text classification. In *NIPS*, 2015.
- [73] Sid Black, Gao Leo, Phil Wang, Connor Leahy, and Stella Biderman. GPT-Neo: Large Scale Autoregressive Language Modeling with Mesh-Tensorflow, March 2021. URL <https://doi.org/10.5281/zenodo.5297715>. If you use this software, please cite it using these metadata.
- [74] Katherine Tian, Eric Mitchell, Huaxiu Yao, Christopher D Manning, and Chelsea Finn. Fine-tuning language models for factuality. In *The Twelfth International Conference on Learning Representations*, 2024. URL <https://openreview.net/forum?id=WPZ2yPag4K>.
- [75] Edward J Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. LoRA: Low-rank adaptation of large language models. In *International Conference on Learning Representations*, 2022. URL <https://openreview.net/forum?id=nZeVKeeFYf9>.
- [76] Tsung-Yi Lin, Michael Maire, Serge Belongie, James Hays, Pietro Perona, Deva Ramanan, Piotr Dollár, and C Lawrence Zitnick. Microsoft coco: Common objects in context. In *Computer vision—ECCV 2014: 13th European conference, zurich, Switzerland, September 6-12, 2014, proceedings, part v 13*, pages 740–755. Springer, 2014.